

# PROG7312 POE Part 1: Task 1 Research

## Five Dashboard User Engagement Strategies Considered:

1. Real-Time Visual Telemetry Feedback – Displaying continuously updating sensor readings through signal traces, status indicators and visual changes allows operators to identify trends, spikes and abnormal behaviour without manually examining raw data.
2. Visual Anomaly Highlighting and Status Indicators – Using distinct visual states for normal, abnormal and disconnected devices allows important problems to immediately attract the operator's attention.
3. System-Level Status Summaries – Presenting key metrics such as the number of connected, reporting, anomalous and silent devices gives users an immediate overview of the overall health of the IoT environment.
4. Interactive Filtering and Drill-Down – Allowing users to filter devices according to categories, locations or alert states reduces information overload and allows operators to investigate a particular device or group of sensors.
5. Proactive Alerts and Notifications – Automatically notifying users when readings exceed expected thresholds, change sharply or stop arriving allows problems to be identified without requiring continuous manual monitoring.

## Chosen Strategy: Real-Time Visual Telemetry Feedback

IoT systems can produce large volumes of heterogeneous telemetry, making it difficult for operators to interpret raw readings quickly. Research into industrial and IoT dashboards shows that visualisation is important because dashboards consolidate sensor observations into a limited number of operational views, allowing users to identify patterns and problems more efficiently [1]. The Smart-X dashboard therefore adopts real-time visual telemetry feedback as its primary user-engagement strategy. This strategy was selected because it directly supports the assignment requirement of helping users identify anomalous spikes and sensor disconnections while monitoring continuously changing data.

The dashboard presents each registered device as a compact telemetry card containing its device name, deployment area, latest reading, recent signal trace and current status. This can be seen in the implemented interface, where devices such as **meter-main**, **meter-pump** and **meter-a2** display recent telemetry visually rather than requiring the operator to interpret a table of raw values. The signal trace provides temporal context: a user can immediately see whether a reading is stable, increasing, decreasing or fluctuating. Research on IoT monitoring similarly identifies interactive graphs and time-series visualisations as useful for understanding individual sensor streams and relationships between devices [2]. Real-time dashboards have also been shown to support early recognition of abnormal trends and system instability when sensor streams and status information are presented together [3].

The strategy is strengthened through visual anomaly highlighting. In the Smart-X interface, sharp changes are represented using orange indicators, while devices with no readings are clearly marked with a red dashed state. This creates an immediate visual distinction between normal and problematic devices. Instead of requiring the operator to inspect every reading, attention is directed towards exceptions. This approach is particularly appropriate for high-throughput IoT environments, where large numbers of sensor observations can otherwise create information overload. Research into real-time IoT dashboards demonstrates the value of displaying active anomalies with identifiable visual states, counters and sensor-specific information so that operators can prioritise unresolved issues [4].

The dashboard also uses summary status cards at the top of the page, displaying the total number of devices, reporting devices, sharp changes, silent devices and total power draw. These indicators provide a rapid system-level overview before the operator examines individual devices. A user can therefore move from an overall problem count to the specific sensor responsible for the issue. This supports a simple troubleshooting workflow: identify an abnormal count, locate the highlighted device, inspect its recent signal trace, and then investigate its deployment or attached configuration information.

Finally, the interface provides continuous feedback rather than relying on a static report. The dashboard displays an updated timestamp and refreshes telemetry, allowing operators to monitor changes as they occur. This aligns with research showing that dashboards can continuously retrieve sensor observations and present them through appropriate visualisations for operational monitoring [1]. Overall, real-time visual telemetry feedback was chosen because it combines immediate system awareness, trend recognition and anomaly identification in a single interface. For Smart-X, this reduces the cognitive effort required to interpret high-volume telemetry and enables users to focus their attention on devices that require investigation.

## References:

- [1] S. Vanden Haute *et al.*, "A Dynamic Dashboarding Application for Fleet Monitoring Using Semantic Web of Things Technologies," *Sensors*, vol. 20, no. 4, p. 1152, 2020.
- [2] M. P. et al., "Detecting anomalies within smart buildings using do-it-yourself internet of things," *Journal of Ambient Intelligence and Humanized Computing*, 2022.
- [3] "IoT-based multi-sensor fusion predictive maintenance of 3D printers using an ensemble model framework," *Discover Computing*, 2026.
- [4] "Real-time IoT architecture for water management in smart cities," *Discover Applied Sciences*, 2024.